

Public Economics and Development

Week 6: Theoretical and empirical evidence on state capacity and weak states

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Drastic differences in capabilities of states across different countries

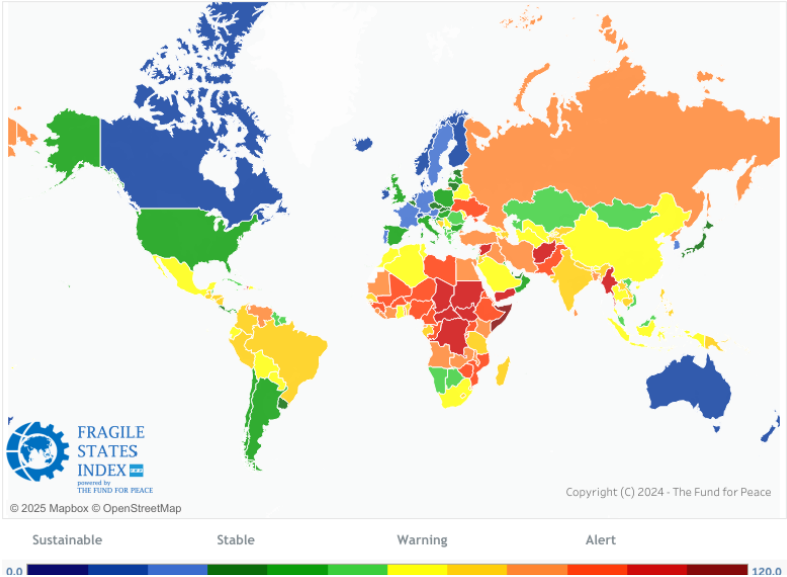
Some of the poorest countries in the world are also 'weak states'

- State capacity: Encompasses ability to support markets, provide public goods, and raise taxes from broad bases
 - ▶ Support markets: Enforce contracts, protect property rights
 - ▶ Provide public goods: Education, health, but also peace (by monopolizing violence)
 - ▶ Broad base taxes (income tax) are more rule-based, less burdensome for individuals, but require large investments for implementation
 - ▶ Narrow base taxes (royalties and tariffs) are easy to implement but could be corruptible and extracts particular group excessively
- Weak states: States that cannot support basic economic function, raise revenues, deliver essential public services, and provide law and order

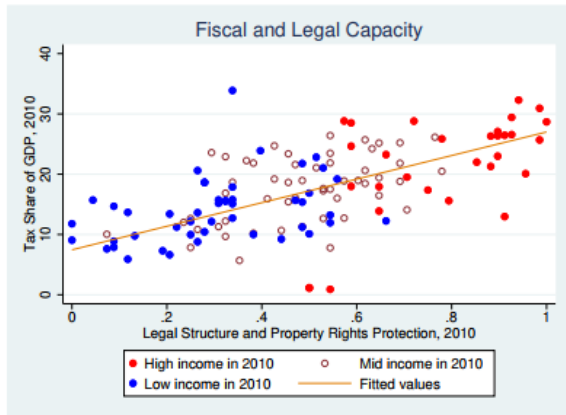
State capacity is widely dispersed, but also clustered

- Weak states are also poorer and mired in internal conflicts
- Developed countries: High income, functioning institutions, less internal conflicts

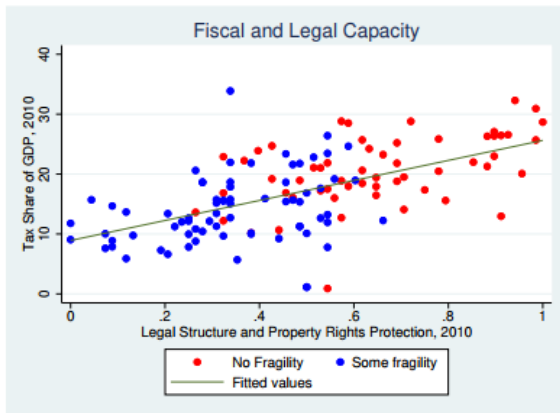
Huge variation in state capacity across the globe



State capacity clustered with income and functioning of institutions



(a) Income and capacities are clustered



(b) Capacities and fragilities index

Why care about state capacity, and open questions?

Why do state capacities matter?

- Capable states can distribute fruits of economic development and prevent violence that stifles the progress
 - ▶ Ability to tax, enforce law, and distribute services contributes to growth and peace
- Ultimately relates to economic and institutional development: Many weak states are poor, authoritarian, and in internal conflict

We discuss open questions on state capacity with theories and empirical findings

- What forces explain the formation of capable states?
- How do weak states perpetuate violence, corruption, and weak growth?
- What explains the clustering of state institutions, violence, and income?

Roadmap for today

1. Overview setup of the topic

2. Conceptualizing state capacity

- 2.1 Besley, Persson (2009): The Origins of State Capacity: Property Rights, Taxation and Politics
- 2.2 Dincecco, Katz (2014): State Capacity and Long-Run Economic Performance
- 2.3 Acemoglu, De Feo, De Luca (2020): Weak States; Causes and Consequences of Sicilian Mafia

3. Reciprocal nature of state capacity

- 3.1 Besley (2020): State Capacity, Reciprocity, and Social Contract
- 3.2 Weigel (2020): The Participation Dividend of Taxation: How Citizens in Congo Engage More With The State When it Tries to Tax Them

Besley, Persson (2009):
The Origins of State Capacity: Property Rights,
Taxation and Politics

Q: What determines the state's investment in fiscal and legal capacity

How are institutional capacity support markets and collect taxes developed?

- Historians note the development of capacity in response to external wars (Tilly 1990)
- Social sciences (economics, especially) typically assume such capacities exist by default
- In reality, institutional capacities differ starkly

This paper provides theoretical framework of state institutions as endogenous investments

- Individuals come from different groups (ruling vs nonruling, different wealth)
- State collects taxes, protects property rights, and provide public goods
- What determines taxation, public goods provision, and support for market activities
- Why are there differences in investments in these capacities across different settings?
 - ▶ Focus on high capacities of wealthy, democratic, and parliamentary democracies
- Provides grounds for studying how institutions evolve during development

Setting up the framework: Definitions and layout

What makes up state capacity

- Legal capacity: Enforce contracts, regulate markets, and protect property rights
- Fiscal capacity: Collect taxes to finance public goods and 'redistribution'

Setting the model up

- There are two time periods $s \in \{1, 2\}$ and two groups of individuals $J \in \{A, B\}$
- Individuals have linear preferences over public goods and post-tax income
- State (led by A or B) protects property rights, collect taxes in both periods
- State can also increase its legal and fiscal capacities at $s = 1$
- State maximizes (politically) weighted sum of individual utilities under budget and capacity constraints

Setting up the framework: Individual utilities

Individual utilities (For given group $J \in \{A, B\}$)

- Gain utility from public good G_s (by factor of α_s) and income Y_s^J (taxed by t_s^J)

$$v_s^J = \alpha_s G_s + (1 - t_s^J) Y_s^J(p_s^J)$$

- Income gain is higher with increase in p_s^J , enforcement of property rights, $\left(\frac{\partial Y_s^J}{\partial p_s^J} > 0\right)$
- Taxes deduct from income but used to finance public goods

Differences across groups

- Demographic share: β^J of population in group $J \in \{A, B\}$ ($\beta^A + \beta^B = 1$)
- Different protection, taxation (negative taxes possible), and income
- Public good is distributed commonly (no differences in level of public goods)
- Assume that group A is ruling at $s = 1$, with chance of continued ruling in $s = 2$ is γ_A

Setting up the framework: State objectives and constraints

Objective of the state: Politically weighted sum of individual utilities

- $\bar{\rho}$ and $\underline{\rho}$: Relative political weights on ruling and nonruling groups ($\bar{\rho}\beta_s^A + \underline{\rho}\beta_s^B = 1$)

$$V_s = \alpha_s G_s + \bar{\rho}\beta_s^A(1 - t_s^A)Y_s^A(p_s^A) + \underline{\rho}\beta_s^B(1 - t_s^B)Y_s^B(p_s^B)$$

- ▶ $\bar{\rho}$ captures how A values its own group relative to demographic share (1 or greater)
- ▶ $\underline{\rho}$ captures how A values nonruling groups (less or equal to 1)
- At $s = 1$, invest in legal (π_s) and fiscal capacity (τ_s) at cost of $L(\pi_2 - \pi_1)$ and $F(\tau_2 - \tau_1)$

Constraints of the state

- Capacity constraints: $p_s^J \leq \pi_s, t_s^J \leq \tau_s$
- Budget Constraints
 - ▶ $s = 1$: Capacity investment included $\rightarrow \sum_J \beta^J t_1^J Y_1^J = G_1 + L(\pi_2 - \pi_1) + F(\tau_2 - \tau_1)$
 - ▶ $s = 2$: No new investment $\rightarrow \sum_J \beta^J t_2^J Y_2^J = G_2$

Policy choices in utilitarian vs politically-motivated settings

Policy choices on p_s^J and t_s^J in utilitarian governments ($\bar{\rho} = \underline{\rho} = 1$)

- p_s^J increases indirect utilities: Provide to the fullest possible $p_s^J = \pi_s$
- t_s^J works in both ways - incomes are taxed, but used for public goods
- $\alpha_s \geq 1$: Tax to the fullest possible ($t_s^J = \tau_s$), all revenues to G_s and investments
- $\alpha_s < 1$: $G_s = 0$, smallest possible taxes (0 in $s = 2$, exact cost of investments in $s = 1$)

Politically-motivated objective maximization ($\bar{\rho} > 1 > \underline{\rho}$)

- No different logic in legal capacity: Highest protection possible
- Public goods are provided only if ruling class values it sufficiently ($\alpha_s \geq \bar{\rho}$)
- If $\alpha < \bar{\rho}$: $G_s = 0$, and taxes could be 'redistributed' from nonruling to ruling class
 - ▶ Ruling group gains $\bar{\rho}$ from 'redistribution', nonruling loses $\underline{\rho}$ from this
 - ▶ With $\bar{\rho} - \underline{\rho} > 0$ beneficial to extract (from nonruling group standpoint)
 - ▶ Implies that in this setting, people have different interests towards fiscal capacity

Investing in state capacity

Investments in fiscal (τ) and legal (π) capacities maximizes expected social utility

- Factors: Wealth (Y_s^J), Value of public goods (α_s), Regime stability (γ_A), representativeness ($\bar{\rho} - \underline{\rho}$)
- Legal capacity is desirable in any case
- When $\alpha_s < \bar{\rho}$, high τ undesirable for non-ruling group $\rightarrow$ Underinvest in capacities
 - ▶ In other cases, no conflict over the desirability of fiscal capacity

Key findings (in theory and empirics)

- In most cases, legal and fiscal capacities complement each other
 - ▶ $\pi \uparrow \rightarrow$ More revenues for public goods $\rightarrow \tau \uparrow \rightarrow$ More funds to increase protection $\rightarrow \pi \uparrow$
- What factors increase investments in state capacities?
 - ▶ High Y_s^J , High α_s , High γ_A , Small $\bar{\rho} - \underline{\rho}$
 - ▶ Mismatch in political power and economic influence $\rightarrow$ Fiscal capacity underinvested

Theoretical prediction matches well with observed correlates

TABLE 1—ECONOMIC AND POLITICAL DETERMINANTS OF LEGAL CAPACITY

	Private credit to GDP (1)	Ease of access to credit (country rank) (2)	Investor protection (country rank) (3)	Index of government anti- diversion policies (4)
Incidence of external conflict up to 1975	0.510*** (0.143)	0.647** (0.191)	0.029 (0.209)	0.576*** (0.170)
Incidence of democracy up to 1975	0.953 (0.059)	0.110 (0.267)	−0.044 (0.078)	0.126** (0.050)
Incidence of parliamentary democracy up to 1975	0.001 (0.063)	0.145 (0.114)	0.339** (0.137)	0.112* (0.061)
English legal origin	−0.009 (0.033)	0.068 (0.057)	0.125** (0.063)	−0.007 (0.040)
Socialist legal origin	—	0.098 (0.111)	0.097 (0.115)	0.010*** (0.035)
German legal origin	0.406*** (0.120)	0.295*** (0.064)	−0.008 (0.149)	0.248*** (0.053)
Scandinavian legal origin	0.112*** (0.041)	0.204*** (0.067)	0.087 (0.098)	0.254*** (0.055)
Observations	93	122	120	115
R ²	0.524	0.334	0.256	0.596

Theoretical prediction matches well with observed correlates

TABLE 2—ECONOMIC AND POLITICAL DETERMINANTS OF FISCAL CAPACITY

	One minus share of trade taxes in total taxes (1)	One minus share of trade and indirect taxes in total taxes (2)	Share of income taxes in GDP (3)	Share of taxes in GDP (4)
Incidence of external conflict up to 1975	0.762*** (0.250)	0.598*** (0.241)	0.579*** (0.220)	0.555*** (0.162)
Incidence of democracy up to 1975	0.143 (0.077)	−0.078 (0.100)	0.091 (0.059)	0.088 (0.059)
Incidence of parliamentary democracy up to 1975	0.031 (0.083)	0.122 (0.103)	0.212*** (0.078)	0.160** (0.068)
English legal origin	−0.038 (0.058)	−0.012 (0.061)	−0.034 (0.043)	−0.015 (0.042)
Socialist legal origin	0.136** (0.058)	−0.222*** (0.037)	−0.109*** (0.065)	−0.119 (0.031)
German legal origin	0.175*** (0.052)	0.196*** (0.090)	0.171* (0.010)	0.010*** (0.083)
Scandinavian legal origin	0.189** (0.077)	0.068** (0.084)	0.258** (0.134)	0.292*** (0.087)
Observations	103	103	103	103
R ²	0.356	0.305	0.600	0.576

Dincecco, Katz (2014):
State Capacity and Long-Run Economic
Performance

Q: long-run relationship btw state capacity and economic growth

Effective states are only a recent phenomenon

- Having enough administrative infrastructure to secure property rights and collect tax
- Modern states underwent (1) fiscal centralization and (2) limited government
 - ▶ Uniform tax systems at national level
 - ▶ Establishment of parliaments that monitor state expenditures

This paper explores relationship between state capacity and economic performance

- Data from European 11 countries from 1650-1913
- Fiscal centralization contribute to long-run economic growth
- Higher extraction of revenues explain how both transformations affect growth

Political transformation

Fiscal centralization: From fragmentation to centralized collection

- Central govt competed with local institutions in tax collection
- Local institutions were free-riding on other localities
- National governments standardized tax implementation (vs bargaining place by place)

Limited government

- Parliaments demanded fiscal oversight to prevent monarchs from wasteful spending
- Having regular control led to rise in extractive capacity

Economic implications: How transformations contribute to growth

- Provision of productive infrastructure (education, public services etc)
- Efficient mobilization of resources (by efficient information gathering and transmission)

Data and regression

Data sources

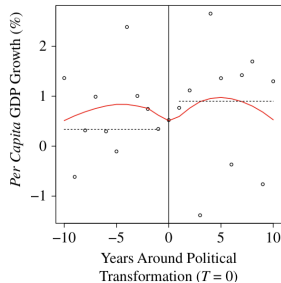
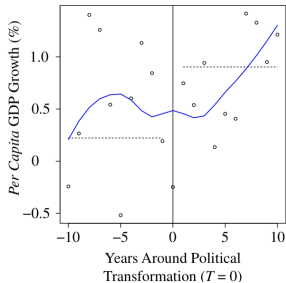
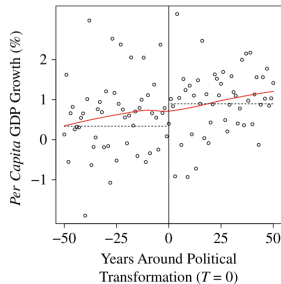
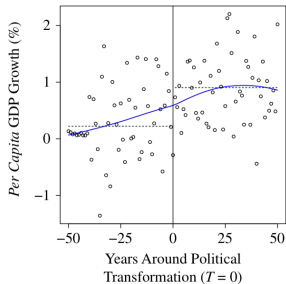
- Apply different timing of fiscal centralization (e.g. ENG: 1066, FRA: 1790, DEN:1903)
- Define timing for limited government (ENG: 1688, FRA: 1870, DEN: 1848)
- State capacity: Per capita national govt revenue and military expenditure
- Economic performance: per capita GDP from Maddison (2010) data
 - ▶ 1600, 1700, then annually from 1820-1913

Identification: Leverage variation in timing of transformations

$$\Delta y_{it} = \alpha + \beta_1 C_{it} + \beta_2 L_{it} + \gamma X_{it-1} + \mu_i + \lambda_t + e_{it}$$

- C_{it} and L_{it} are fiscal centralization and limited government dummy
- X_{it-1} : External/internal conflicts, population growth, and lags of dependent variables
- Country and year fixed effects included
- Potential reverse causality concerns (why lags of dependent variable are in X_{it-1})

Growth in economic performance after political transformations



Effects are larger for fiscal centralizations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable is real <i>per capita</i> GDP growth							
Fiscal centralisation	0.657 (0.087) (0.000)	0.222 (0.086) (0.041)	0.268 (0.125) (0.033)	0.166 (0.094) (0.137)	0.249 (0.099) (0.030)	0.292 (0.116) (0.063)	0.344 (0.171) (0.063)
Limited government	0.321 (0.109) (0.027)	0.053 (0.165) (0.757)	-0.028 (0.146) (0.837)	-0.102 (0.230) (0.728)	-0.108 (0.113) (0.385)	0.049 (0.225) (0.836)	-0.080 (0.205) (0.738)
Fiscal centralisation × limited government				0.232 (0.239) (0.433)	0.311 (0.126) (0.010)		
Lag (1) <i>per capita</i> GDP growth						-0.185 (0.094)	-0.195 (0.094)
Lag (2) <i>per capita</i> GDP growth						-0.168 (0.056)	-0.178 (0.053)
R ²	0.049	0.202	0.207	0.202	0.055	0.244	0.254
Observations	1,772	1,772	1,772	1,772	1,772	1,750	1,750

Notes. Estimation method is OLS. All specifications include country fixed effects. Columns (2)–(4) and (6)–(7) include year fixed effects and column (5) includes a linear time trend. Additionally, columns (3) and (7) include country-specific time trends. Cluster-robust standard errors clustered by country in parentheses (first line below coefficients) and p-values for two-sided Wald tests computed according to Cameron *et al.*'s (2008) wild bootstrap-t procedure also in parentheses (second line below coefficients for Fiscal centralisation, Limited government and interaction term).

Extractive capacity increases explains the findings

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Dependent variable is <i>per capita</i> revenue growth								
Fiscal centralisation	1.405 (0.497) (0.001)	1.467 (0.531) (0.007)	2.929 (0.926) (0.039)	3.254 (0.738) (0.025)	6.610 (2.526) (0.108)	3.770 (1.251) (0.028)	3.923 (0.907) (0.006)	4.268 (1.081) (0.035)
Limited government	0.438 (0.564) (0.466)	0.047 (0.638) (0.973)	0.717 (0.652) (0.348)	1.215 (0.631) (0.120)	1.883 (0.758) (0.017)	1.209 (0.989) (0.254)	1.601 (0.807) (0.080)	5.117 (2.791) (0.236)
External conflict dummy (lagged)		-0.774 (1.237)	-0.777 (1.115)	-1.257 (0.508)	0.620 (1.180)	0.957 (1.887)	-0.540 (0.692)	1.526 (1.796)
Internal conflict dummy (lagged)		2.697 (0.858)	3.008 (0.962)	1.854 (1.587)	4.190 (1.518)	2.106 (1.131)	1.262 (1.823)	0.499 (2.495)
Population growth (lagged)		-0.191 (0.110)	-0.192 (0.110)	-0.064 (0.090)	-0.226 (0.056)	-0.274 (0.171)	-0.134 (0.149)	-0.256 (0.162)
Lag (1) <i>per capita</i> revenue growth						-0.173 (0.026)	-0.169 (0.028)	-0.227 (0.030)
Lag (2) <i>per capita</i> revenue growth						-0.132 (0.049)	-0.128 (0.044)	-0.201 (0.057)
Lag (3) log <i>per capita</i> revenues								-11.401 (2.405)
R ²	0.160	0.162	0.165	0.008	0.176	0.197	0.045	0.235
Observations	1,760	1,748	1,748	1,748	1,019	1,734	1,734	1,734

Notes. Estimation method is OLS. All specifications include country fixed effects. All specifications include year fixed effects except for columns (4) and (7), which include linear time trends. Additionally, columns (3)–(8) include country-specific time trends. The sample period is 1650–1845 for column (5). Cluster-robust standard errors clustered by country in parentheses (first line below coefficients) and p-values for two-sided Wald tests computed according to Cameron *et al.*'s (2008) wild bootstrap-t procedure also in parentheses (second line below coefficients for Fiscal centralisation and Limited government).

Takeaways and conclusions

Key findings

- Fiscal centralization (and limited govt, to some extent) contribute to long-run growth
- Particularly driven by rise in extractive capacity
- The findings implies that growth of administrative infrastructure had significant role

Remaining questions

- Potential role for non-fiscal mechanisms?
 - ▶ Fiscal centralization removes political power of local elites (e.g. feudal landlords)
 - ▶ Political accountability: Prevent wasteful spending and have voice in policy decisions
- How are taxes collected used is also a remaining question
 - ▶ Efficient bureaucracy that provides services at efficient costs?

Acemoglu, De Feo, De Luca (2020):
Weak States; Causes and Consequences of
Sicilian Mafia

Q: How do illicit organizations emerge and contribute to weak states?

Criminal organizations are widespread and may play a significant role in economic growth

- Think of Mafias in Italy, drug cartels in Mexico/Colombia, and so on
- Many of these have filled the void left by weak states
 - ▶ Some of these groups have provided public services on their own
 - ▶ But also collected 'taxes' in exchange for 'protection'
- Possibly contribute to weakness of the state and economic underdevelopment?

This paper explores the growth and consequence of Sicillian Mafia

- What explains Mafia's growth dating back to 1860s?
- How does proliferation of Mafia affects economic development and other outcomes?
 - ▶ Literacy rates and high school completion rates (provision of education)
 - ▶ Possible mechanism: Capture (literal and figurative) of politicians and candidates

Proliferation of Mafia in Sicily

Sicilian Mafia origins

- Mafia started during the transition from Bourbon Kingdom to Unified Italy in 1860s
- Unified Italy not fully set up in Sicilian Islands → Weak state presence
- Rise of socialist movement (*Fasci*), especially in the severe droughts of 1893
- This movement led landowners and local politicians turn to Mafia for protection

Impact of the drought of 1893

- Conditions for agriculture were deteriorating even prior to 1893
- 1893: Drop in crop production → Fasci movement → landowners turn to Mafia
- This pattern is confirmed from data obtained by the authors

Mafia and Draught distribution significantly overlap

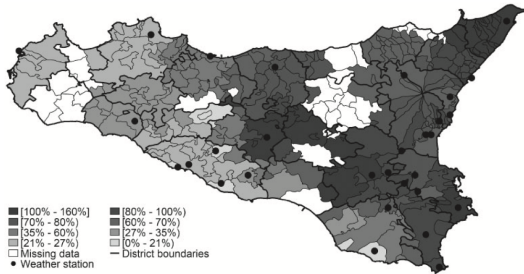


FIGURE 2

Drought intensity in spring 1893. Ratio of the rainfall in spring 1893 to long-run average spring rainfall.

(a) Draughts in Sicily

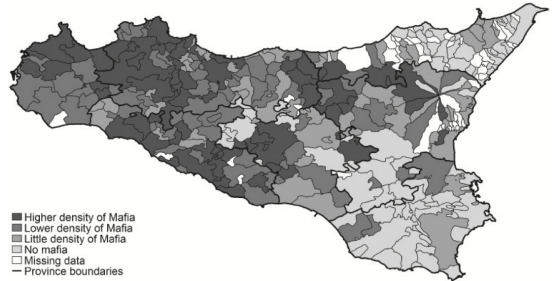


FIGURE 3

Mafia intensity in 1900 according to the Police inspector Cutrera. Source: [Cutrera \(1900\)](#).

(b) Mafia presence in Sicily

Data and regression

Database

- Mafia presence: Obtained from Cutrera (1900)
- Fasci presence and rainfall: Renda (1977) and Eredia (1918), respectively
- Human capital outcome: Obtained from Italian census (1911, 21, 31, 61, 71, 81)
- Other public goods: Various mortality rates and expenditure from ISTAT

Identification leverages exogenous variation in Mafia presence explained by draughts

- Reduced form: $y_i = \alpha + \beta \text{Mafia}_i + \gamma X_i + e_i$
- IV (first-stage): $\text{Mafia}_i = \alpha_1 + \beta_1 \text{rainfall}_i + \gamma_1 X_i + u_i \Rightarrow \widehat{\text{Mafia}}_i = \hat{\alpha}_1 + \hat{\beta}_1 \text{rainfall}_i + \hat{\gamma}_1 X_i$
- IV (second-stage): $y_i = \alpha + \beta \widehat{\text{Mafia}}_i + \gamma X_i + e_i$
- Relevance: F statistics around 10 (not very strong, but not weak either)
- Exclusion: Droughts only affect outcomes through Mafia presence

Significant mid-term effect on human capital

TABLE 9
The impact of Mafia on Human Capital

Dependent variable	Literacy in 1911			Literacy in 1921			Literacy in 1931		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: IV results									
Mafia 1900	-0.07 (0.04)	-0.06 (0.04)	-0.09 (0.05)	-0.10 (0.05)	-0.09 (0.04)	-0.10 (0.05)	-0.07 (0.04)	-0.06 (0.03)	-0.08 (0.04)
Panel B: OLS results									
Mafia 1900	0.00 (0.01)	-0.00 (0.01)	-0.00 (0.01)	0.01 (0.00)	0.00 (0.00)	0.00 (0.01)	-0.00 (0.01)	-0.01 (0.01)	-0.00 (0.01)
R^2	0.26	0.27	0.32	0.28	0.29	0.32	0.35	0.36	0.43
Province FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Determinants of Fasci	✓	✓	✓	✓	✓	✓	✓	✓	✓
Determinants of Mafia		✓	✓		✓	✓		✓	✓
Geographic controls			✓			✓			✓
Observations	245	245	245	245	245	245	245	245	245

Findings robust to other measures of public service provision

Dependent variable	Infant mortality in 1909			Development Expenditure 1912			Notaries in 1924			HHI in 1909		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: IV results												
Mafia 1900	0.04 (0.02)	0.04 (0.02)	0.05 (0.02)	-2.40 (1.10)	-2.08 (0.93)	-1.45 (0.84)	-0.12 (0.07)	-0.11 (0.06)	-0.12 (0.06)	0.41 (0.18)	0.34 (0.15)	0.31 (0.14)
Panel B: OLS results												
Mafia 1900	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.05 (0.15)	0.01 (0.15)	0.02 (0.14)	-0.02 (0.01)	-0.02 (0.01)	-0.02 (0.01)	0.03 (0.02)	0.04 (0.02)	0.03 (0.02)
R^2	0.28	0.31	0.36	0.10	0.10	0.24	0.10	0.12	0.16	0.22	0.24	0.30
Province FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Determinants of Fasci	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Determinants of Mafia		✓	✓		✓	✓		✓	✓		✓	✓
Geographic controls			✓			✓			✓			✓
Observations	243	243	243	245	245	245	245	245	245	242	242	242

Long-run negative (but weaker) effects on human capital

	1961			1971			1981		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: High school									
Mafia 1900	-0.02 (0.01)	-0.02 (0.01)	-0.01 (0.01)	-0.03 (0.01)	-0.02 (0.01)	-0.02 (0.01)	-0.04 (0.02)	-0.03 (0.02)	-0.02 (0.02)
Panel B: Literacy									
Mafia 1900	-0.04 (0.02)	-0.03 (0.02)	-0.04 (0.02)	-0.03 (0.02)	-0.02 (0.01)	-0.03 (0.02)	-0.02 (0.01)	-0.01 (0.01)	-0.01 (0.01)
Panel C: Infant mortality									
Mafia 1900				0.00 (0.01)	0.00 (0.01)	0.00 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)
Panel D: Development expenditure									
Mafia 1900	-0.56 (0.40)	-0.50 (0.34)	-0.02 (0.36)						
Panel E: Aqueduct availability									
Mafia 1900	-0.13 (0.10)	-0.12 (0.08)	-0.17 (0.09)	-0.13 (0.07)	-0.10 (0.06)	-0.07 (0.04)	-0.14 (0.08)	-0.12 (0.08)	-0.09 (0.08)
Panel F: HHI									
Mafia 1900	0.10 (0.04)	0.08 (0.03)	0.09 (0.02)	0.08 (0.04)	0.07 (0.03)	0.06 (0.03)	0.09 (0.03)	0.07 (0.03)	0.05 (0.03)
Province FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Determinants of Fasci	✓	✓	✓	✓	✓	✓	✓	✓	✓
Determinants of Mafia		✓	✓		✓	✓		✓	✓
Geographic controls			✓			✓			✓
Observations	245	245	245	245	245	245	245	245	245

Besley (2020): State Capacity, Reciprocity, and Social Contract

Q: How does civic culture contribute to fiscal capacity?

Increased tax intake enabled governments to spend on essential services

- Not just on national defense, but also health care, education, income transfers etc
- Switches in public spending requires economic, political, and social changes
 - ▶ Should develop capacity to tax and raise revenues
 - ▶ But also, cultivate civic culture that increases tax *compliance*

This paper extends the Besley, Persson (2009) model

- By incorporating the role of civic culture, reciprocity, and social contract
- Relative payoffs of civic-minded citizens and materialistic citizens matter
- Investigates how civic culture can develop in the long-run
- Ultimately answers how state capacity can persist with interactions with citizens

Motivating findings: Different attitudes to tax compliance

DETERMINANTS OF ATTITUDES TOWARDS TAX COMPLIANCE^a

Low confidence in government	–	–0.177(0.029)	–
Low trust in people	–	–	0.009 (0.022)
Male	–0.212 (0.018)	–0.210 (0.018)	–0.212 (0.018)
Age 30–49	0.222 (0.021)	0.220 (0.021)	0.222 (0.021)
Age 50+	0.571 (0.043)	0.564 (0.043)	0.572 (0.043)
Education: Middle	0.042 (0.026)	0.046 (0.026)	0.042 (0.026)
Education: Upper	0.097 (0.032)	0.101 (0.033)	0.098 (0.031)
Income 2	–0.017 (0.055)	–0.014 (0.029)	–0.017 (0.055)
Income 3	–0.034 (0.054)	–0.032 (0.054)	–0.034 (0.053)
Income 4	–0.087 (0.056)	–0.087 (0.057)	–0.087 (0.056)
Income 5	–0.043 (0.056)	–0.041 (0.057)	–0.043 (0.056)
Income 6	–0.141 (0.056)	–0.141 (0.057)	–0.141 (0.056)
Income 7	–0.161 (0.057)	–0.160 (0.057)	–0.161 (0.057)
Income 8	–0.158 (0.069)	–0.157 (0.069)	–0.158 (0.069)
Income 9	–0.194 (0.063)	–0.191 (0.062)	–0.193 (0.063)
Income 10	–0.234 (0.081)	–0.236 (0.080)	–0.233 (0.081)
Number of observations	303,229	303,229	303,229

^aThe dependent variable is based on question “Is it justifiable to cheat on your taxes if you have a chance?” from the World Values Survey and European Values Survey (various waves) with the scale reversed so that the highest score is associated with cheating not being justified. All specifications include wave and country dummies with standard errors clustered at the country level. The data cover

Source: Individual responses from World Value Survey, regressing compliance on individual characteristics, country and wave dummies. Income is displayed from lowest to highest

Model setup

Basic structure of the model

- Share e of elites who decides public goods and transfers, and non-elites
- Among non-elites, μ_s of civic-minded citizens and $1 - \mu_s$ of materialists
 - ▶ All subject to taxes, with possible noncompliance (n), and receive utility from public goods
 - ▶ Civic-minded: Gains if tax used on public goods (G), loss if used on transfers to elites (B)
 - ▶ Noncompliance: Incurs cost $C(n)$, which rises with fiscal capacity τ (easily detected)
 - ▶ Individual Utility: $\lambda > 0$ for civic-minded citizens, $\lambda = 0$ for materialists

$$\alpha G + b + w(1 - (1 - n)[t - \lambda\{G - eB\}] - \tau C(n))$$

- Revenues must be spent on public goods G , transfers to elites (B) and non-elites (b)
 - ▶ For every B , share $0 \leq \sigma \leq 1$ must be spent on b . Thus $b = \sigma B$ (Higher $\sigma \rightarrow$ cohesiveness)
 - ▶ Budget: $eB + (1 - e)\sigma B = T - G$
- Fiscal capacity is determined by physical investments but also 'tax morale'
 - ▶ For materialists, only physical investments matter
 - ▶ For civic-minded person, public good $\uparrow \rightarrow$ tax morale $\uparrow$ (transfer $\uparrow \rightarrow$ tax morale $\downarrow$)

Optimal policy choice

Optimal policy choice

- Elites choose t , G , and B to maximize the sum of individual utilities
- Subject to cohesiveness of institutions (σ), fiscal capacity (τ) and civic culture (μ)
- Whether revenues are spent on G or B depends on cohesiveness and civic culture
 - ▶ Public goods are more likely to be provided in scenario where civic culture ($\mu \uparrow$) and cohesiveness ($\sigma \uparrow$) are dominant (and also raise fiscal capacity)
 - ▶ This comes from the fact that civic-minded persons are more likely to comply if revenues are used for public goods (and vice versa)
 - ▶ Furthermore, high cohesiveness makes transfers to elites expensive $\rightarrow$ make public goods relatively cheaper
- Over time, additional public goods due to $(\mu, \sigma) \uparrow \Rightarrow$ benefits of being civic-minded $\uparrow$
 - ▶ Conversely, narrow range of public goods provision from $(\mu, \sigma) \downarrow \Rightarrow$ civic-mindedness $\downarrow$
- Initial state of civicness, cohesiveness + strength of common interest (α) matters

Core implications of the model

Cohesiveness of institutions and civic culture is complementary

- Cohesiveness of institutions are strengthened when more civic-minded persons induce high public goods over transfers to elites
- People are more likely to be civic minded if institutional setups make transfers to elites costly (through cohesiveness)

These determine how common interests develop

- In a less cohesive, weakly civic-minded setting, external threat may not significantly increase common interest

Civic-mindedness and cohesiveness can be fostered in more representative political setup

- Elites that more closely represents non-elites more likely to sustain power
- Further increases returns for being civic-minded

Potential extensions

High dependence on outside sources of funds

- Outside sources of funding reduces need for own-source tax revenues
- This reduces importance of tax compliance and the role of civic culture
- Sudden, exogenous changes to outside funds → Challenges in raising fiscal capacity

Can elites determine civic-mindedness (μ)?

- We assumed that μ is exogenously determined
- But these can be potentially determined by elites
- Room for extension: Making tax avoidance costly, Benefits of public services over time

Takeaways: How these topics can be empirically studied

How can fiscal capacities be built

- Evolution towards broad-based system (Gordon, Li 2009; Jensen 2022)
- Efforts to gather more accurate information in developing countries (Balan et al 2021)

What increases tax morale and compliance?

- Increase benefits of compliance (tax breaks for formal firms) - (Benhassine 2018)
- Tax perceptions and electoral outcomes (Christensen, Garfias 2021)

Other factors that contribute to state capacities

- Role of bureaucrats: Well-motivated, well-trained bureaucrats matter for effective public goods provision
- Active studies on how to recruit and monitor them (Dal Bó et al 2013)

Weigel (2020):
The Participation Dividend of Taxation: How
Citizens in Congo Engage More With The State
When it Tries to Tax Them

Q: Do taxation increase civic engagement in fragile states?

Participation dividend of taxation

- Taxes enable state to provide public goods and services
- It is said to also stimulate political engagement: Taxation as a social contract
 - ▶ Traditionally: Citizens delay paying taxes until rulers made concessions
- Does this form of bargaining also appear in fragile states?

This paper uses randomized rollout of tax campaigns to see if taxes raises participation

- Context: Democratic Republic of Congo
- Rollout of tax campaign meant to increase citizens registering to pay taxes
- Active visit of tax collectors in treated towns vs. Declaration system in control group
- Participation & perceptions that government are responsible for public goods ↑↑

Institutional context of DRC

Kangara, DRC

- DRC is one of the poorest country in the world (188/200 in terms of tax-GDP ratio)
- Tax revenue per person at Kangara: \$0.23
- Very few people pay taxes to begin with: 6% of population have ever paid taxes

Tax collection and political participation

- Systematic tax collection began in 2016, induced by redistricting of the region
- Citizens have limited avenue for political participation
- Neighborhood meetings, letter of complaint are main channels

Research design

Randomized control treatment

- Door-to-door property tax collection campaign in Apr-Dec 2016
 - ▶ Property registration and tax solicitation
- Treated neighborhoods (253) received this in 2016, (control (178): mid-2017)
- Stratified randomization based on location and population

Estimation leverages random assignment to treatment controlling for strata

$$y_{ijk} = \beta I_{jk} + \alpha_k + X_{ijk}\gamma + X_{jk}\phi + e_{ijk}$$

- i : individual, j : neighborhood, k : strata
- Control for individual and neighborhood characteristics (X_{ijk} and X_{jk})
- Control & treated groups balanced on observables and tax/political outcome variables

Tax compliance rises following treatment

EFFECTS OF THE CAMPAIGN ON COLLECTOR VISITS, TAXPAYER REGISTRATION,
PROPERTY TAX COMPLIANCE, AND REVENUES

Dependent variable:	Visited by collector	Registered as taxpayer	Property tax compliance		Tax revenue per person
Unit:	Household (1)	Household (2)	Household (3)	Neighborhood (4)	Neighborhood (5)
Campaign	0.815*** (0.013)	0.788*** (0.009)	0.103*** (0.007)	0.115*** (0.009)	367.295*** (62.518)
Stratum FE	Yes	Yes	Yes	Yes	Yes
R^2	0.640	0.577	0.054	0.396	0.173
Observations	27,443	27,443	27,443	356	356
Clusters	356	356	356	N/A	N/A
Control mean	0.0499	0.0000	0.0006	0.0005	1.5683

Citizens' political participation increases

	Town hall meeting attendance (1)	Evaluation form submission (2)	Town hall or evaluation (3)	Town hall and evaluation (4)	Index (town hall & evaluation) (5)	Cost of participation (transport) (6)	Cost of participation (transport & opp.) (7)
Campaign	0.045** (0.020)	0.024** (0.012)	0.050*** (0.016)	0.027*** (0.009)	0.145*** (0.043)	0.050*** (0.017)	0.071*** (0.021)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stratum FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.068	0.055	0.071	0.039	0.073	0.054	0.058
Observations	1,934	2,913	2,913	2,913	2,913	2,913	2,913
Clusters	252	356	356	356	356	356	356
Control mean	0.17	0.099	0.16	0.035	-0.077	0.11	0.16
Dep. var.	Binary	Binary	Binary	Binary	Std. index	% Daily inc.	% Daily inc.
Rand. inf. p	.023	.058	.0048	.0048	.0022	.0072	.0022
Bonferroni p	.033	.067	N/A	N/A	N/A	N/A	N/A

Citizen's perception of government responsibility rises

EFFECTS OF THE CAMPAIGN ON CITIZENS' BELIEFS ABOUT THE GOVERNMENT

Dependent variable	$\hat{\beta}$	Std. err.	R^2	N	μ_c
Panel A: Responsibility of government					
Resp. for public goods provision	0.117**	0.051	0.041	2,913	-0.063
Panel B: Extractive capacity of government					
Information about taxpayers	0.147***	0.044	0.086	2,910	-0.075
Ability to punish tax evaders	0.046	0.047	0.044	2,883	-0.027
Perceived citizen tax compliance	0.343***	0.053	0.100	1,954	-0.170
Performance of tax ministry	0.122***	0.047	0.065	2,791	-0.104
Panel C: Productive capacity of government					
Ability to provide public goods	-0.013	0.053	0.038	2,484	0.007
Performance of government	0.043	0.049	0.042	2,795	-0.048
Panel D: Integrity of government					
Integrity of tax collectors	0.187***	0.044	0.043	2,732	-0.136
Integrity of government spending	0.109**	0.050	0.054	2,766	-0.100
Transparency of government	0.031	0.045	0.078	2,890	-0.042

Takeaways and remaining questions

Effects of tax collection campaign

- Higher tax compliance (registered properties, willingness to pay taxes)
- Higher demand for political participation (esp. town halls)
- Higher recognition of state responsibility and expectation on state's roles

Remaining questions

- Political participation follows tax collection
 - ▶ Governments willing and able to respond to demands will proceed as is
 - ▶ Otherwise, may attempt other ways to raise revenue (e.g. seignorage, tariffs)
- Other ways to increase political participation and civic engagement?
 - ▶ What about people who do not pay taxes (i.e. most people in developing countries)
 - ▶ How do demand from citizens shape politician behavior?

Unanswered questions in differences in taxation across countries

Effectiveness of the state is not a given

- Many states are unable to mobilize sufficient resources
- Which affects their ability to provide essential public services
- Also have welfare consequences for the citizens
- What ways to increase capacity?
 - ▶ Developing countries differ in underlying institutional conditions
 - ▶ Many have underdeveloped financial systems and income registries and bureaucracies

Raising extractive capacities must be balanced with citizen demand

- Collect taxes → citizens demand essential services and state responsibility
- States unable to meet these demands induce tax non-compliance
- So how do institutional trust and state capacities interact?